TECHNICAL SPECIFICATION

TECHNICAL SPECIFICATION FOR 30 V, 75 AH STATIONARY BATTERIES

1(i). Scope : The specification covers the design, manufacture, testing at the Manufacturer's works, delivery by road transport to different sites of West Bengal State Electricity Distribution Company Limited of Lead Acid Battery set suitable for Auto and Manual Float cum Boost Battery Charger with 30 V DC Output. Supervision and erection/commissioning of devices shall have to be undertaken by mutual acceptance of terms and conditions for the same, if required.

APPLICATION:

The system requires a reliable and uninterrupted D.C. supply for supplying D.C. Power to emergency lights, closing and tripping coils of circuit breakers, relays, semaphores etc.

1(ii). Service conditions:

Equipment to be supplied against this specification shall be suitable for satisfactory continuous operation under the following tropical conditions.

Maximum ambient temperature (Degree C)	50
Maximum temperature in shade (Degree C)	45
Minimum Temperature (Degree C)	3
Relative Humidity (percent)	95
Maximum Annual rain fall (mm)	1450
Maximum wind pressure (kg/sq. m)	150
Maximum altitude above mean sea level (Meter)	1500
Isoceraunic level (days per year)	50
Seismic Level	The sites fall within seismic zone-III and IV as classified in the IS:1983
Moderately hot and humid tropical climate conductive to rust and fungus growth	

1(iii). Standards:

Unless otherwise specified, the equipment shall conform to latest applicable Indian standard of equivalent IEC, British or USA standard and in particular to the following standard (or equivalent IEC British, USA standard):

The tenderer shall clearly state the standard to which the equipment offered by him conforms.

IS: 1652/ 1991	**Specification for stationary lead acid Plante Cell battery**

<u>TECHNICAL SPECIFICATION FOR LEAD ACID BATTERIES (30 V, 75 AH)</u>

1.1 Low maintenance type of Lead Acid stationary Batteries incorporating of pure Lead Lamellar type with plante formation positive plates assembled in 2 Volt containers with a capacity of 75 Ampere–hour at 10 hour rate of discharge to an end voltage of 1.85 V per Cell having Electrical performance as per IS:1652/1991 or its latest version amended upto date.

1.2 The terminals of the battery shall be suitable for receiving cable lugs. 15 Nos. of 2 volts 75 AH Cell connected in series shall form one set of 30 volts 75 AH.

2.0 Application : The device shall be installed in indoor location within the battery Room of Substations to provide auxiliary DC Power Supply for following applications:

a) Emergency lighting

b) Circuit breaker/switchgear/protective relay operations

c) Equipment supervision indication

d) Audio visual annunciation

Under normal conditions, auxiliary DC Power Supply for continuous load will be met from Battery Charger and storage batteries should be kept under float or boost charging condition.

But in case of Circuit Breaker/Switchgear/Protective Relay operation and/or in the event of failure of incoming power supply to Battery Charger, required auxiliary DC Power supply shall be met from Storage Batteries.

3.0 STANDARD RATING: The standard ratings of batteries shall be 30 Volts DC, 75 AH at 27 Deg C. Duty cycle is to be mentioned.

4.0 Cell Voltage : The nominal voltage of a single cell shall not be less than 2 volts at the beginning of Charging.

The trickle charging voltage per cell shall be within 2.25 volts to 2.3 volts.

The boost charging voltage per cell shall be upto 2.75 volts.

At the end of the charging, cell shall be floated easily around 2.25 volts without causing adverse corrosion or loss of water.

5.0 Charging Current:

Trickle charging current shall be between 60 mA to 180 mA.

Boost charging current shall be within following limits:-

a) Starting Rate : 10.5 Amp

b) Finishing Rate : 5.25 Amp.

Fully charged battery shall accept about 1 mA/AH or less current under float application.

6.0 The expected life span of the batteries shall be minimum 10 years.

7.0 Constructional Feature:

7.1 Type: The storage batteries shall consist of 15 nos. Plante type Lead Acid Cells of Voltage not less than 2 volts per cell at the beginning of initial charging.

7.2 Main Components of each cell

7.2.1 Positive Plates

The positive plates of cell shall be constructed from pure lead consisting of laminations.

7.2.2 Negative Plates

The negative plates shall be pasted type consisting of sturdy lead alloy grid filled with active material.

7.2.3 Plate Connections

The plug of the plates of like polarity shall be connected to terminal post for external connections. Suitable plastic buffer, moulded bottom plate support shall be used for holding plates in proper position.

7.2.4 Separator

The separators shall be constructed from highly micro-porous and rigid material of appropriate shape. Separators shall be inert chemically and shall prevent internal short circuit and shall ensure minimum internal resistance.

7.2.5 Terminal Posts

The positive and negative terminal posts shall be clearly and indelibly marked for easy identification. The terminal posts shall be sealed properly with the lid by rubber grommets or other suitable device. The terminal posts shall have provision for inter-cell/outgoing cable connections.

7.2.6 Container

Transparent styrene acryl nitrile (SAN) container for each cell shall be made of sturdy construction having proven excellent clarity, outstanding chemical resistant property, rigidity with high insulating quality. The container shall provide permanent transparency to enable visual inspection of electrolyte level and internal condition of cell. Recommended electrolyte level shall be clearly and indelibly marked.

7.2.7 Cell Lid

Cell lid for covering cell container shall be made from high quality non-corrosive plastic materials and shall have provision for easy removal.

7.2.8 Vent Plug

A vent plug of suitable design shall be provided on each cell lid. The vent plug shall have a micro-porous plastic alloy/ceramic filter which will prevent escape of acid fume/spreading of acid fume but will allow free exit of oxygen and hydrogen generated in the process of charging.

7.2.9 Electrolyte

The electrolyte for the cell shall be battery grade sulphuric acid conforming to IS:266-1977 or its latest version amended upto date and diluted with distilled water to specific gravity 1.2 at 27 Deg C. The lead acid cell batteries shall be supplied in dry and uncharged condition. Diluted sulphuric acid of approved quality and required quantity shall be supplied in separate non-returnable porcelain or any other acid and corrosive proof jars. 10% extra electrolyte shall have to be supplied.

7.3 Other components of Battery

7.3.1 Cell pillars and connectors :

Cell Pillars and connectors shall be made from highly conductive material of lead allow having generous cross section ideally suited for high current duties.

Bolts, nuts, washers etc. for connecting the cells and connectors shall be coated with anti-corrosive highly conductive material.

7.3.2 Battery Stand

The stand for supporting battery cells shall be provided. The stand for battery cell shall be of Mild Steel finished with at least three coats of anti-sulphuric paints of approved shade. The racks shall be free standing type. Suitable porcelain/plastic insulators shall be provided between the stand and the battery room floor. Number plate to designate each cell of battery shall be provided and shall be attached on the rack. Provision shall be made for clamping outgoing cable on the rack. The cell shall be supported on the rack in suitable row and tier formation with adequate clearance between adjacent cells.

7.3.3 Polarity Marking

The polarity marking of the terminals shall be marked for identification. The positive terminal may be identified by **"P"** or **(+)** sign or **red** colour mark and the negative terminal may be identified by **"N"** or **(-)** or **blue** colour. Terminal marking shall be permanent and non-deteriorating.

7.3.4 Life

The Bidder shall quote in his offer the guaranteed life of the battery when operating under the conditions specified in the specification.

7.3.5 Battery shall be transported in dry, uncharged condition. The electrolyte shall be supplied separately in non-returnable container of acid resistant material. Each battery set shall be supplied with operation / commissioning manual.

7.3.6 The following characteristics have to be satisfied by the batteries offered:

i) Loss of capacity on storage of a fully charged battery for 28 days shall not be more than 3%.

ii) Ampere hour efficiency shall not be less than 97 %.

iii) Distilled water addition once in 6 months should be sufficient to maintain the electrolyte level.

iv) The battery should withstand conditions of under-floating and over-floating.

v) The battery should be capable of being maintained at a higher Electrolyte specific gravity of 1.230 +/- 0.005 without deterioration to Grid corrosion.

vi) The Bidder should furnish sufficient evidence of his capability to manufacture the low maintenance batteries.

7.3.7 Test on Battery:

All tests shall be carried as per the relevant standard. Test shall comprise of Routine tests and Acceptance Tests.

7.3.8 Routine Test :

These are to be performed at manufacture's works and shall include the following test:-

i) Verification of constructional requirement.
ii) Verification of marking.
iii) Verification of dimensions.
iv) Test for capacity.
v) Test for voltage during discharge.
vi) Ampere hour and watt – hour efficiency test.
vii) Test for loss of capacity on storage.
viii) Endurance test.

7.3.9 Acceptance Test:

These tests shall be performed in presence of purchaser's representative from the sample selected from the lot offered for supply.

i) Verification of constructional requirement.
ii) Verification of marking.
iii) Verification of dimensions.
iv) Test for capacity.
v) Test for voltage during discharge.
vi) Ampere hour and watt – hour efficiency test.

7.3.10 Important design criteria

Plate connectors and paste shall be designed to contribute maximum effective surface area, maximum electrical conductivity and superior voltage characteristics throughout service life. The plates shall be designed for maximum performance durability and shall not buckle during different service condition i.e. high rate of discharge and rapid fluctuation of load. The container should have provision to avoid short-circuit of plates due to deposition at the bottom.

7.3.11 Accessories

Following accessories in addition to battery shall have to be supplied in each set of equipment:

Sl. No.	**Item Description**	**Qty**
01	Syringe type Hydrometer set complete in all respect and capable of indicating specific gravity reading in steps of 0.005 for Plante type Lead Acid Cell Battery	1 no.
02	Cell Testing Voltmeter (Range: 3-0-3 volts having scale conforming to IS)/ Digital Multimeter	1 no.
03	Mercury Glass Thermometer Range: 0-100 Deg C having suitable scale representing 1Deg C temperature rise.	1 no.
04	Wall mounting Plastic holder for holding 1 no. Hydrometer and 1 no. Thermometer.	1 no.
05	Plastic syringe: 10 Oz(ounce)	1 set
06	Plastic Funnel : 150 mm dia.	1 no.
07	Acid resistant plastic Jug: 2 ltrs. Capacity	1 no.
08	Rubber Siphon: 12.7 mm dia 2 mtr long	1 no.
09	Rubber apron	2 nos.
10	Rubber gloves	2 Pairs
11	Rubber boots: Knee height	2 Pairs
12	Special tools	AS required
13	Battery Stand of Mild Steel	1 no.
14	Inter battery connector of lead plated copper of cross section not less than 50 sq. mm.	14 Nos. +2 extra
15	Lead plated set containing 2 nuts,1 bolt and 2 washers	30 sets+4 extra
16	Sulphuric Acid (10% extra) of 1.190 Specific Gravity at 27°C sufficient for first filling of the Battery	1 Lot

8.0 Testing Facilities :

The Bidder must clearly indicate what testing facilities are available in the works of manufacturer and whether the facilities are adequate to carry out all Routine, Acceptance Tests. These facilities should be available to WBSEDCL's Engineers, if deputed to carry out or witness the tests in the manufacturer's works. If any of the tests can not be carried out in the manufacturer's works, Bidder shall have to arrange for such testing at any NABL Accredited/Govt. Test house or Laboratory at his own cost.

9.0 Inspection :

All tests and inspection shall be made at the place of manufacturer unless otherwise specially agreed upon by the manufacturer and the WBSEDCL. The manufacturer shall provide the WBSEDCL all reasonable facilities, without charge to satisfy him that the material is being supplied in accordance with this specification.

10.0 Tests:-

General: The equipment including all components and accessories shall be subjected to all type of tests including Routine and acceptance tests in accordance with provision contained in relevant standard.

10.1 Type Test: The Bidder shall have to submit along with their Tender documents , as pre-requisites, the complete type Test Reports as stipulated in the relevant IS/IEC, carried out within 5 years from the due date of Tender, from CPRI/NABL accredited/Govt. recognised Test House or Laboratory on the offered Item, failing which their offer may not be technically acceptable.

10.2 Routine and acceptance Tests: Routine &acceptance tests shall have to be carried out in compliance with provision contained in the relevant standard and / or to ascertain satisfactory performance of the offered device at the works of the Manufacturer.

The acceptance tests shall have to be conducted in the presence of authorised representative of the purchaser before effecting delivery.

11.0 Drawings :

The successful bidder shall have to submit the 6(six) copies of final drawings, manuals, literatures for approval to the office of the Chief Engineer (P & E) for approval before starting manufacture of the equipment. Before despatch of the equipment, 6(six) copies of drawings, manuals and literature shall be submitted to purchaser for distribution to different offices of the Company.

In addition to above, every crate of complete set of equipment shall also contain in waterproof folder, 2(two) sets of drawing, manuals and literature for commissioning, operation and maintenance at site.

12.0 PERFORMANCE GUARANTEE:

Battery shall be delivered to the various consignees of WBSEDCL and shall be suitably packed to avoid damages during transit. The Battery with all its integral part will be guaranteed for the period of 5 years from the date of last dispatch.

In the event of any defect in any of its integral part of the equipment arising out of faulty design, materials, workmanship within the above period, the supplier shall guarantee to replace or repair the same to the satisfaction of the purchaser. However, any engineering error, omission, wrong provision, etc. which do not have any effect on the time period, shall be attended to as and when observed/pointed out without any price implication.

13.0 GUARANTEED TECHNICAL PARTICULARS

The bidder should fill up the details in schedule A – 'Guaranteed Technical Particulars' and the statement such as "as per drawing enclosed", "as per WBSEDCL requirement", "as per IS", "as per specification" etc. shall be considered as details not furnished and such offers may be rejected.

14.0 Schedules:

The Bidder shall fill in the following schedules, which is part and parcel of the tender specification and offer. If the schedules are not submitted duly filled in with the offer, the offer shall be liable for rejection.

Schedule A: Guaranteed Technical Particulars

Schedule 'A'

GUARANTEED TECHNICAL PARTICULARS FOR PLANTE TYPE 30V, 75 AH LEAD ACID CELL BATTERY

Sl No.	Particulars	Details		
1.	Name and Address of the Manufacturer			
2.	Application standards			
3.	Type & Designation as per standards			
4.	Manufacturer's type and designation			
5.	Capacity and voltage at 27◦C a) At 10hrs. rate of discharge b) At 5hrs. rate of discharge c) At 3 hrs. rate of discharge d) At 1 hrs. rate of discharge e) At 1 minute rate of discharge	AH	WH	FINAL VOLTAGE (v)
6.	Cell details a) No. of cells per battery b) No. of positive plates per cell c)Total no. of plates per cell: d) Type of positive plate: e) Type of negative plate: f) Surface area of plate in sq.mm: g) Construction details and dimensions of i) Positive plate ii) Negative plate			

	h) Rated current of each positive plate: i)Construction details of separators including thickness, type and material j) Container details: i) Material of container ii)Overall dimensions of container in mm : Length : Width : Height k) Overall dimension of cell in mm (including Cell height) l) Weight of each of cell in kg. a) Without Acid b) With acid m) Clearance in mm between: i) Top of plates and top of container ii) Bottom of plates & bottom of container ii) Edges of plates & inner surfaces of container n) Cell Lid details: i) Material ii) Type iii) Feature	
7.	Electrolyte details: i) Applicable standard of electrolyte	

	ii) Quantity of Electrolyte & Specific gravity at 27◦C for first filling in each cell iii) Quantity of Electrolyte required for 15 nos. Plante type lead acid cell plus 10% extra iv) Specific gravity of electrolyte at 27 ◦C with all cells fully charged v) Specific gravity of electrolyte at the end of 10 hour of discharge rate		
8.	Cell Pillars &Connectors : i) Material ii) Type iii)Dimension details in mm iv) Details of inter row inter tier connectors including bolts, nuts and washers.		
9.	Battery Stand: 1) No. of racks per battery 2) No. of cells per rack 3) Formation of row and tier details 4) Material of stand 5) Particulars of anti-sulphuric paint to be provided 6) Type and no. of stand insulator to be provided		
10.	Nominal cell voltage in volts:		
11.	Recommended rate of first charging battery cells 1) Current in amps 2) Voltage in volts	**Start**	**Finish**

	3) Total minimum input during initial charging in AH		
12.	Recommended float charge rate		
13.	Recommended float charge voltage across the battery terminals		
14.	Recommended boost charge voltage across the battery terminals		
15.	Guaranteed AH efficiency at 10 hr. rate of discharge		
16.	Guaranteed WH efficiency at 10 hr. rate of discharge		
17.	Internal Resistance of charged cell in milli ohm		
18.	Resistance of the charged battery including inter connector between the cell in ohms		
19.	Short Circuit Current for dead short circuit across the battery terminals when 1) Battery in floating mode 2) Battery in boost charge mode		
20.	Cell voltage characteristics curves during charging at 0.5,1.0 and 1.5 times normal rate		
21.	Battery layout arrangement		
22.	List of accessories to be provided with battery		

Date: Signature:

Place: Name:

Name of the Company:

Seal of the Company: